

NUTRITION AND FLUID THERAPY

ARAM ABDULLAH



Learning objectives

- ☐ To understand the importance of perioperative nutritional and fluid balance
- ☐ The nutritional requirement of surgical patients and the effect of intestinal resection on nutrition
- ☐ The causes and complications of malnutrition and their management
- ☐ The fluid component and balance
- ☐ Option of nutritional intervention and the indication for enteral versus parenteral nutritional support



INTRODUCTION

Aim: Identify patients at the risk of malnutrition and to ensure nutritional requirements.

Malnutrition occurs in

about 30% of surgical patients with GI disease and about 60% of those in which hospital stay has prolonged due to postoperative complications



PHYSIOLOGICAL RESPONSES TO NUTRITIONAL IMPAIRMENT

1. Metabolic Response to fasting or starvation



STARVATION(FIRST 12 HOURS)

Liver-Organ of glucose production

Brain tissue , RBCs ,WBCs ,Renal medulla –utilize only glucose for their metabolic needs

Muscle glycogen----undergo glycogenolysis----
Lactate----Glucose(Cori cycle){takes place in liver}



FASTING(>24HOURS)

Glycogen become depleted

De novo glucose production(gluconeogenesis) starts in liver

Sources: Non carbohydrate precursors like catabolism of
glutamine and alanine

It can use up to 75g of skeletal muscle per day



WITH MORE PROLOGED FASTING

Increased dependence on fat oxidation

Fat stores undergo increased breakdown to form----
Glycerol----glucose

Hepatic production of ketones from fatty acids



48-72 HRS OF FASTING

CNS uses ketone bodies as primary fuel source(fat fuel economy)

This has reduced the need for muscle breakdown by about 55g
per day



Significant reduction in REE

It is mediated by decline in conversion of T4 to T3

Obligatory glucose requirement -200g/dl under conditions of
prolonged fasting



Metabolic response to starvation

- Low plasma insulin
- High plasma glucagon
- Hepatic glycogenolysis
- Protein catabolism
- Hepatic gluconeogenesis
- Lipolysis: mobilisation of fat stores (increased fat oxidation) – overall decrease in protein and carbohydrate oxidation
- Adaptive ketogenesis
- Reduction in resting energy expenditure (from approximately 25–30 kcal/kg per day to 15–20 kcal/kg per day)



METABOLIC RESPONSE TO TRAUMA AND SEPSIS

Impaired formation of ketones

Breakdown of protein to synthesis glucose

Cannot be entirely prevented by glucose



catabolic response to trauma and sepsis

- Increased counter-regulatory hormones: adrenaline (epinephrine), noradrenaline (norepinephrine), cortisol, glucagon and growth hormone
- Increased energy requirements (up to 40 kcal/kg per day)
- Increased nitrogen requirements
- Insulin resistance and glucose intolerance
- Preferential oxidation of lipids
- Increased gluconeogenesis and protein catabolism
- Loss of adaptive ketogenesis
- Fluid retention with associated hypoalbuminaemia



NUTRITIONAL ASSESSMENT

ABCD

- ✓ **Anthropometry** - parameters include BMI, mid arm circumference & skinfold thickness.
- ✓ **Biochemistry-** Albumin, CRP & WBC ~ markers of inflammation, which may compromise nutritional status.
 - ✓ Hb level ~ indicates anemia ~ lack of vitamins.
 - Glycated hb ~ DM
 - Sodium & urea reflects the renal function
 - Ca & phosphate ~ anticipation of refeeding syndrome.



- ✓ **Clinical evaluation** - upper abdominal symptoms like nausea, vomiting, early satiety, dysphagia, bloating and lower gi symptoms like diarrhoea, constipation indicates of some sort of inadequate nutritional intake or absorption.

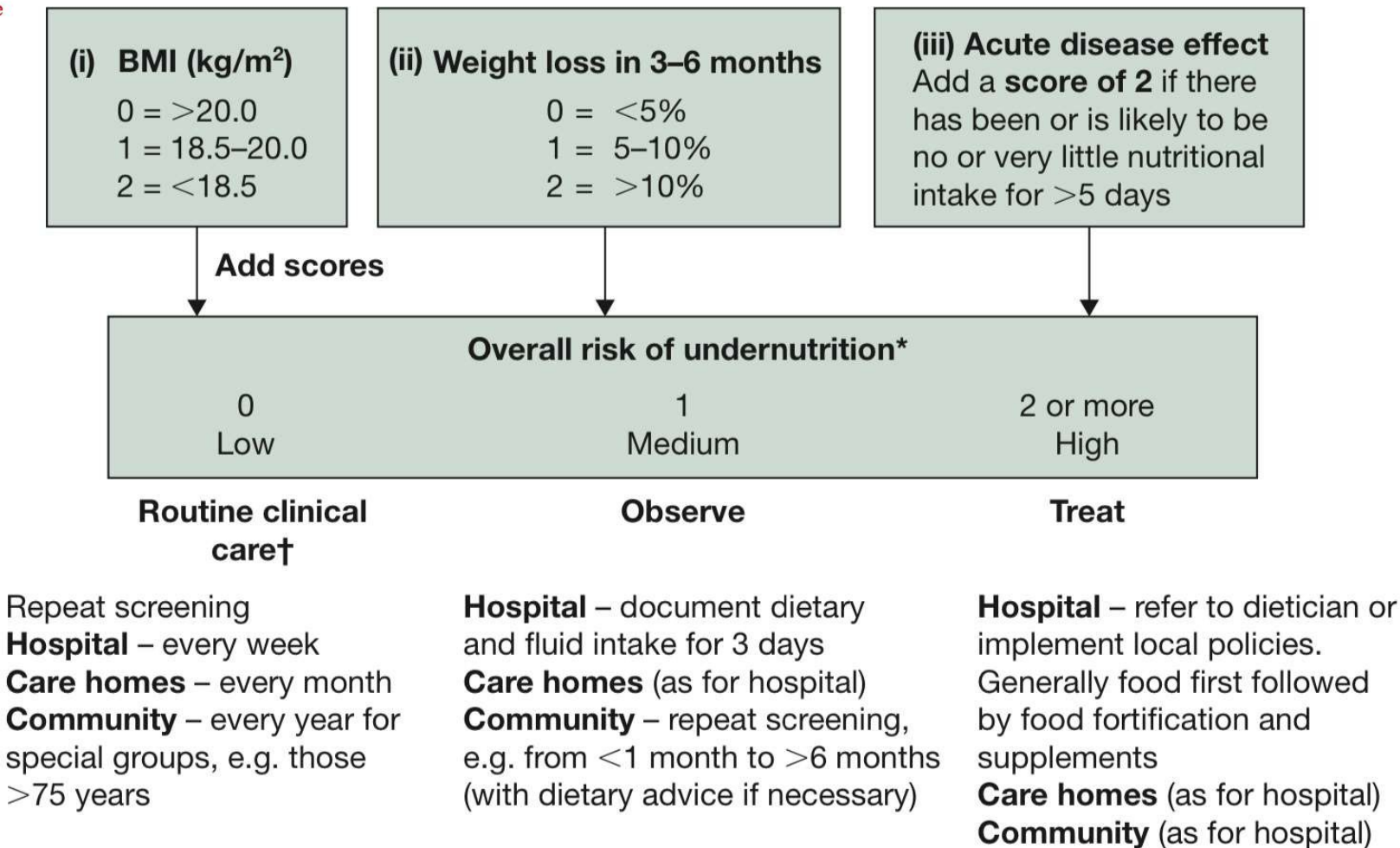
Enquiry about past medical histories and comorbidities.(cancer,IBD,liver diseases, stroke, dementia,...)

- ✓ **Dietary assessment tools** - 24 hour recall method

Malnutrition universal screening tool(MUST) - rapid screening tool , can be used both in hospitals and community.



The MUST tool



FLUID AND ELECTROLYTE REPLACEMENT

Daily fluid balance

Fluid intake = liquid ingested as oral fluids + fluid released during oxidation of consumed foods

TABLE 25.1 Estimated daily fluid balance for a healthy 70-kg adult in a temperate climate.

Intake (L)		Output (L)	
Water from beverages	1.2	Urine	1.5
Water from food	1.0	Insensible losses (skin and lungs)	0.9
Metabolic processes of oxidation	0.3	Faeces	0.1

Loss increases in conditions like pyrexia, exertion on warm environments, also patients with tracheostomy (insensible loss), diarrhoea, chronic bowel pathergies (fecal route).

Prescriptions of fluid should aim to restore the fluid losses and also provide adequate water & electrolytes. To maintain the ICF & ECF.

Healthy individual of 70 kg requires;

2.2 liter per day : water

Sodium: 0.9-1.2 mmol/kg/day

Potassium: 1 mmol/kg/day

Calcium: 5mM/day

Mg: 1mM/day



IV fluid replacement solutions

If any GI absorptive impairment or large fluid losses - IV fluids are indicated.

Commonly used : 0.9% saline, Hartmann's solution & 5 % dextrose.

Colloid fluids can also be used.(Gelofusine, Volplex, Voluven).

None of the different iv fluids mentioned above have electrolyte levels that completely mirror plasma levels, thus there is no single ideal fluid replacement therapy.

Type of fluid replacement therapy is determined by individual patient's need. (depends on the nature of fluid loss and amount of fluid replacement necessary).



Assessment includes pulse, BP, central venous pressure.

Accurate input and output charts

Measurement of serum electrolytes and hematocrit.

Choice also depends on the type of GI fluid loss , as composition of GI secretions varies with anatomical location.



NUTRITIONAL REQUIREMENTS

WHEN PLANNING A FEEDING REGIMEN, THE PATIENT SHOULD BE WEIGHED
AND
AN ASSESSMENT MADE OF DAILY ENERGY AND PROTEIN REQUIREMENTS.
IT IS ESSENTIAL TO MONITOR DAILY INTAKE TO PROVIDE AN ASSESSMENT
OF TOLERANCE.

Macronutrient requirements

Energy

The total energy requirement of a stable patient with a normal or moderately increased need is approximately 20–30 kcal/kg.

In the majority of hospitalized patients in whom energy demands from activity are minimal, total

energy

requirements are approximately 1300–1800 kcal/day.per day.



CARBOHYDRATE

It form 45 to 65% of the total calorie intake per day

Optimal utilization of energy during nutritional support is ensured by avoiding the infusion of glucose at rates approximating physiological maximums.

FAT

Dietary fat is composed of triglycerides of predominantly four long-chain fatty acids.

Safe and non-toxic fat emulsions based upon long-chain triglycerides (LCTs) have been commercially available for over 30 years.



Protein

The basic requirement for nitrogen in patients without pre-existing malnutrition and without metabolic stress is 0.10–0.15 g/kg per day.

In hypermetabolic patients, the nitrogen requirements increase to 0.20–0.25 g/kg per day.



VITAMINS, MINERALS AND TRACE ELEMENTS

THE VITAMIN C REQUIREMENT INCREASES TO 60–80 MG/DAY.

The water-soluble vitamins B and C act as coenzymes in collagen formation and wound healing.

Supplemental vitamin B12 is often indicated in patients who have undergone intestinal resection or gastric surgery and in those with a history of alcohol dependence.

Absorption of the fat-soluble vitamins A, D, E and K is reduced in steatorrhoea ,

potassium and phosphate are all subject to significant losses, particularly in patients with diarrhoeal illness.

Their levels need daily monitoring and appropriate replacement



UP TO 50 PER CENT OF THE SMALL INTESTINE CAN BE SURGICALLY REMOVED OR BYPASSED WITHOUT PERMANENT DELETERIOUS EFFECTS. BILE SALTS ARE ESSENTIAL FOR THE ABSORPTION OF FATS AND FAT-SOLUBLE VITAMINS. THE ENTEROHEPATIC CIRCULATION OF BILE SALTS IS CRITICAL TO MAINTAIN THE BILE salt pool following resection

With extensive resection (<150 cm of remaining small intestine), metabolic and nutritional consequences arise, resulting in the disease entity known as short bowel syndrome



Small bowel motility

Small bowel motility is three times slower in the ileum than in the jejunum. In addition, the ileocaecal valve may slow transit.



ileum

The ileum is the only site of absorption of vitamin B12 and bile Acid salts are essential for the absorption of fats and fat-soluble vitamins.

The enterohepatic circulation of bile salts is critical to maintain the bile salt pool.

Following resection of the ileum, the loss of bile salts increases and is not met by an increase in synthesis. Depletion of the bile salt pool results in fat malabsorption. In addition, loss of bile salts into the colon affects colonic mucosa, causing a reduction in salt and water absorption, which increases stool losses.



colon

Transit times in the colon vary between 24 and 150 hours. The efficiency of water and salt absorption in the colon exceeds 90 per cent. Another important colonic function is the fermentation of carbohydrates to produce short-chain fatty acids.



Effects of resection

Resection of proximal jejunum results in no significant alterations in fluid and electrolyte levels as the ileum and colon can adapt to absorb the increased fluid and electrolyte load.

Resection of ileum results in a significant enhancement of gastric motility and acceleration of intestinal transit.

Following ileal resection, the colon receives a much larger volume of fluid and electrolytes and it also receives bile salts, which reduce its ability to absorb salt and water, resulting in diarrhea

Regular parenteral vitamin B12 is required.



Short bowel syndrome

short bowel syndrome is who have had in excess of 200 cm of small bowel resected together with colectomy

They are conveniently divided into two groups termed 'net absorbers' and 'net secretors'.

Absorbers

characteristically have more than 100 cm of residual jejunum and they absorb more water and sodium from the diet than passes through the stoma.

These patients can be managed without supplementary parenteral fluids.

Secretors

usually have less than 100 cm of residual jejunum and lose more water and sodium from their stoma than they take by mouth.

These patients require supplements. Their usual daily jejunostomy output may exceed 4 litres per 24 hours.



Complication of short bowel syndrome

peptic ulceration related to gastric hypersecretion

cholelithiasis because of interruption of the enterohepatic cycle of bile salt

hyperoxaluria as a result of the increased absorption of oxalate in the colon predisposing to renal stones.

slurred speech, ataxia and altered affect

Treatment

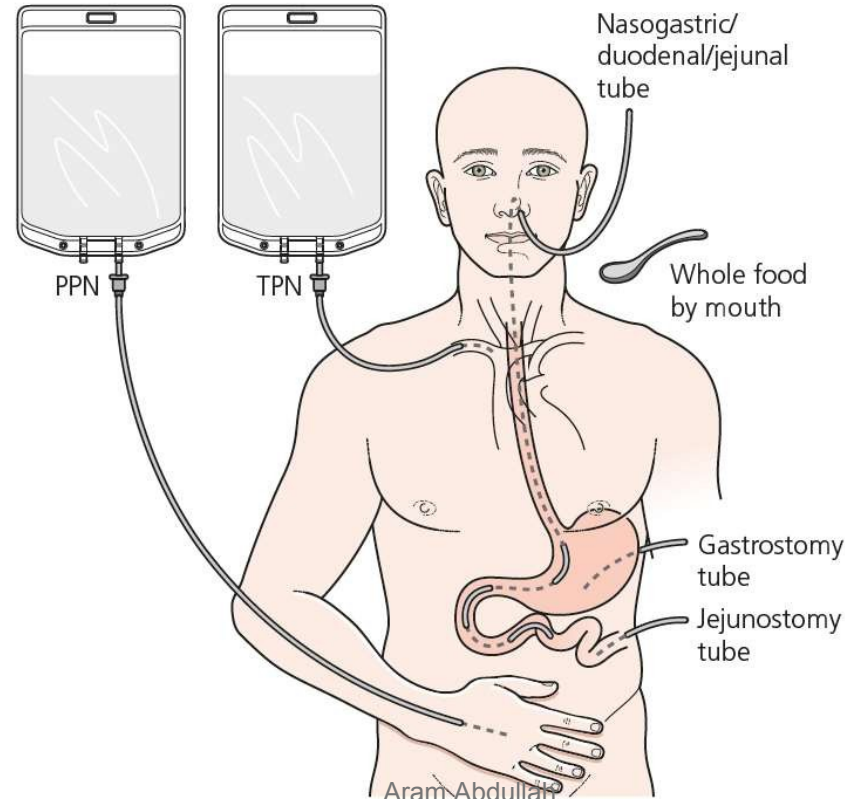
Low carbohydrate diet.

Anti-secretory drugs which include H₂-receptor antagonists, proton pump inhibitors and the somatostatin analogue octreotide.

Octreotide also reduces gastrointestinal motility.

Anti-motility drugs include loperamide and codeinephosphate

ARTIFICIAL NUTRITIONAL SUPPORT





ENTERAL NUTRITION

Delivery of nutrients into the gastrointestinal tract

The alimentary tract should be used whenever possible.

This can be achieved with oral supplements (sip feeding) or with a variety of tube-feeding techniques delivering food into the stomach, duodenum or jejunum.



BENEFITS OF ENTERAL NUTRITION

Preservation of gut mucosal barrier

Prevention of gut atrophy

Reduced infection rate

Reduced hospital stay

Better wound healing



A variety of nutrient formulations is available for enteral feeding.

Polymeric feeds which contain intact protein.

Monomeric/elemental feeds contain nitrogen in the form of free amino acids or, in some cases, peptides.

Newer feeding formulations which contain glutamine and fibre to optimise intestinal nutrition or immunonutrients such as arginine and fish oils.



SIP FEEDS

Sip feeds are used in patients who can drink but whose appetites are impaired or in whom adequate intakes cannot be maintained.

These feeds typically provide 200 kcal and 2 g of nitrogen per 200 mL carton. This sip feeding technique increase calorie and nitrogen intakes without detriment to spontaneous nutrition.



Tube feeding techniques

- ☐ Nasogastric tube
- ☐ Fine bore feeding tube
- ☐ surgical or percutaneous endoscopic gastrostomy (PEG)
- ☐ post-pyloric feeding utilising nasojejunal tubes

Conventionally, 20–30 mL are administered per hour initially, gradually increasing to goal rates within 48–72 hours.

In most units, feeding is discontinued for 4–5 hours overnight to allow gastric pH to return to normal.

All tubes should be flushed with water at least twice daily.

If a build up of solidified diet occurs instillation into the tube of agents such as chymotrypsin or papain may salvage a partially obstructed tube.

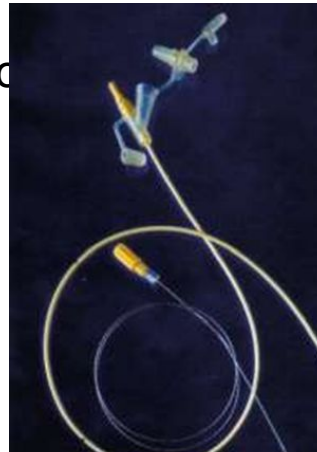


Fine-bore tube insertion

The patient should be semi-recumbent. The introducer wire is lubricated and inserted into the fine-bore tube. The tube is passed through the nose and into the stomach via the nasopharynx and esophagus. The wire is withdrawn and the tube is taped to the patient. There is a small risk of malposition into a bronchus or of causing pneumothorax. The position of the tube should be checked using plain abdominal radiography.

Alternatively, 5 mL of air can be injected and a stethoscope used to confirm bubbling from the stomach.

Confirmation of position by pH testing is possible but limited by the difficulty of obtaining a fluid aspirate with narrow lumen tubes.





Gastrostomy

The placement of a tube through the abdominal wall directly into the stomach is termed 'gastrostomy'. Today, the majority are performed by percutaneous insertion under endoscopic control using local anesthesia, known as PEG (percutaneous endoscopic gastrostomy) tubes

Two methods of PEG are available

1. DIRECT STAB TECHNIQUE

Endoscope is passed and the stomach filled with air. The endoscopist then watches a cannula entering the stomach having been inserted directly through the anterior abdominal wall. A guidewire is then passed through the cannula into the stomach. A gastrostomy tube may then be introduced into the stomach through a 'peel away' sheath.



2. TRANSORAL OR PUSH-THROUGH TECHNIQUE

a guidewire or suture is brought out of the stomach by the endoscope after transabdominal percutaneous insertion and is either attached to a gastrostomy tube or the tube is pushed over a guidewire.

The abdominal end of the wire is then pulled, advancing the gastrostomy tube through the esophagus and into the stomach. Continued pulling abuts it up against the abdominal wall.

If patients require enteral nutrition for prolonged periods(4–6 weeks), then PEG is preferable Necrotising fasciitis and intra-abdominal wall abscesses are most complications



Jejunostomy

Achieved using nasojejunal tubes or by placement of needle jejunostomy at the time of laparotomy.

Associated with a reduction in aspiration or enhanced tolerance of enteral nutrition.

Nasojejunal tubes often necessitate the use of fluoroscopy or endoscopy to achieve placement, which may delay commencement of feeding.



COMPLICATIONS OF ENTERAL FEEDING

Tube related

Malposition

Displacement

Blockage

Breakage/leakage

Local complications (e.g. erosion of skin/mucosa)



Gastrointestinal

Diarrhoea

Bloating, nausea, vomiting

Abdominal cramps

Aspiration

Constipation

Metabolic/biochemical

Electrolyte disorders

Vitamin, mineral, trace element deficiencies

Drug interactions



PARENTERAL NUTRITION

TPN is indicated in patients who are unable to meet their nutritional requirements via absorption of nutrients from their intestinal tract.

This can be either in addition to enteral feeding (supplemental parenteral nutrition) or the sole source of nutrition (TPN).

The commonest cause for this is in patients with short bowel syndrome related to massive intestinal resection or a significant reduction in functional small bowel, often related to intestinal fistulation.

In some cases ,TPN is a temporary for a few days to minimize nutritional depletion until a route of enteral nutrition is established

COMPOSITION

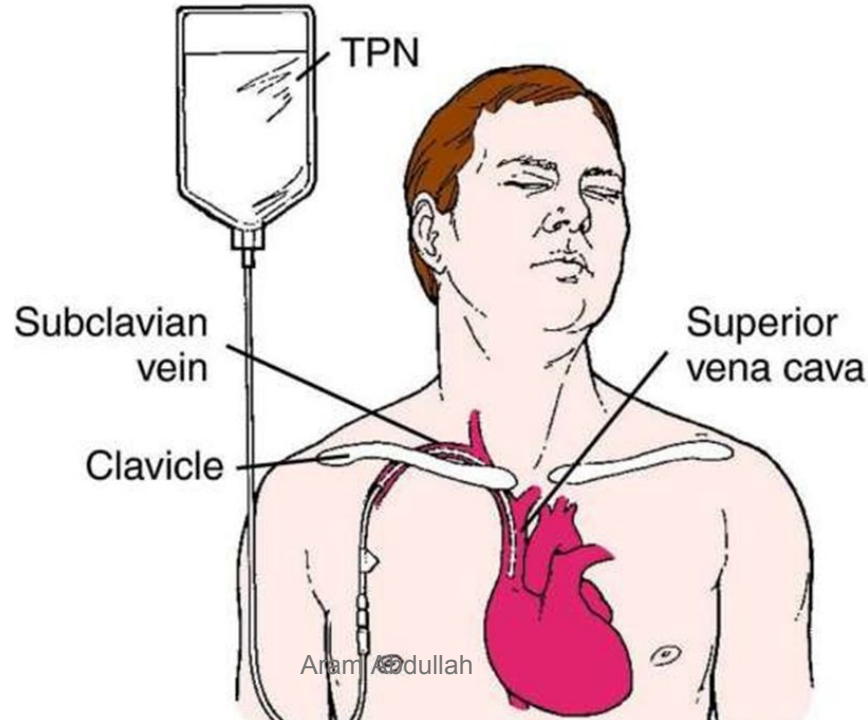
- ❖ 3-litre bag containing a lipid emulsion with a mixture of essential and non-essential amino acids, glucose, electrolytes, trace elements and vitamins.
- ❖ The energy content is in the ratio of 150–250 kcal per gram of protein nitrogen, with usually 30–50% of the energy coming from fat.
- ❖ Folic acid is supplemented in the solution once or twice a week at a dose of 15 mg and other vitamins are given daily.
- ❖ Patients requiring long-term parenteral nutrition (over months) given a single-dose injection of vitamin B12.
 - ❖ 20–30 mmol phosphate is required daily to ensure phosphorylation of glucose and prevent hypophosphatemia.



ADMINISTRATION

- Administered directly into the central venous system (the superior vena cava or the right atrium) to minimize the risk of venous thrombophlebitis, through either a peripherally inserted central catheter (PICC) or central venous catheter.
- PICC lines may be inserted through the basilic, cephalic, brachial or median cubital vein of the arm
- Femoral lines should be avoided because of the high risk of infection at this site.

Post radiographs should be performed after PICC or central venous catheters are inserted to confirm the correct position of the line tip within the SVC or right atrium prior to commencing the parenteral feed





- ✓ The parenteral nutrition bag should be covered at all times, including during infusion, with an opaque protective bag to prevent the vitamins from degradation.
- ✓ If the parenteral nutrition infusion is disconnected from the line for any reason during administration the bag will need to be discarded.
- ✓ It contributes to fluid intake, and thus the volume infused should be carefully recorded on the fluid balance chart to avoid fluid overload.
- ✓ Parenteral nutrition withdrawal planned in a stepwise manner and stopped once the patient is established on adequate oral or enteral support.



COMPLICATIONS OF PARENTERAL FEEDING

Insertion complications

- Pneumothorax
- Misplacement

Line complications

- ☐ Sepsis
- ☐ Thrombosis

Metabolic complications

- Electrolyte disorders, including refeeding syndrome
Blood sugar derangement
 - Liver dysfunction
- Metabolic bone disease
 - Vitamin deficiencies



Two of more of the following:

BMI <18.5 kg/m²

Unintentional weight loss $>10\%$ within the last 3–6 months

Little or no nutritional intake for more than 5 days

History of alcohol abuse or on medication, including insulin,
chemotherapy, antacids or diuretics

Underlying pathological process is hypophosphataemia, resulting in fluid and electrolyte shifts between the intra- and extracellular compartments.

They may develop arrhythmias, muscle weakness, respiratory or cardiac failure, oedema, lethargy or seizures; at its most severe the syndrome can be fatal.



PREVENTION AND THERAPY

Correct electrolyte abnormalities before starting nutrition support.
Continue to monitor serum electrolytes after nutrition support begins.
Initiate nutrition support at low rate/concentration and advance to goal slowly in patients who are at high risk.

THANK YOU